

CLASH OF THE **CLAWS**

RVCE_SynthShield

Artificial User

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Idea

Problem Statement

Why this theme? On-Device AI · Productivity · GenAI Research

Samsung races to ship Galaxy AI features, but every test needs real user data. GDPR, Knox, and Zero Trust block it — so AI ships undertested.

What problem are we solving?

Modern AI systems require massive datasets for training and validation, but real-world data in healthcare, finance, education, and enterprise systems is restricted by GDPR, HIPAA, and Zero Trust policies. Existing synthetic data tools generate statistically weak or unrealistic records that fail to simulate real-world AI scenarios effectively.

Who faces this? (persona)

AI devs at Samsung · Privacy & compliance teams · GenAI researchers · Fintech & healthcare builders

Why is it important?

Every undertested AI day costs user trust. GDPR violations cost millions. A privacy-safe test layer isn't nice-to-have — it's mandatory.

Current Solution & Gaps

Existing solutions (if any)

1. Anonymised real data — re-identification risk still exists, blocked by Zero Trust.
2. Basic synthetic data tools — random fields, not real behavioural patterns.
3. Manual test scripts — don't scale, miss user diversity.

What's missing / broken

No tool today gives data that is privacy-guaranteed, behaviourally realistic, AND pluggable into a live AI test pipeline — all three at once.

The Opportunity

Build an agent that fabricates complete fake humans with real-feeling digital footprints that AI can safely test against.

Current Solution & Gaps

OUR SOLUTION :

SynthShield is a synthetic intelligence engine that generates behaviourally realistic, privacy-preserving datasets for AI training and testing.

The platform combines:

- Differential Privacy
- Generative AI
- Schema-aware data synthesis
- AI-ready export pipelines

to safely create scalable synthetic datasets without exposing sensitive real-world information.

Your Solutions

What we built

SynthShield — a 3-layer autonomous agent that generates synthetic user personas and populates a 'Data Tank' with fake emails, calendar events, and app logs.

Core idea explained

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touching real data.

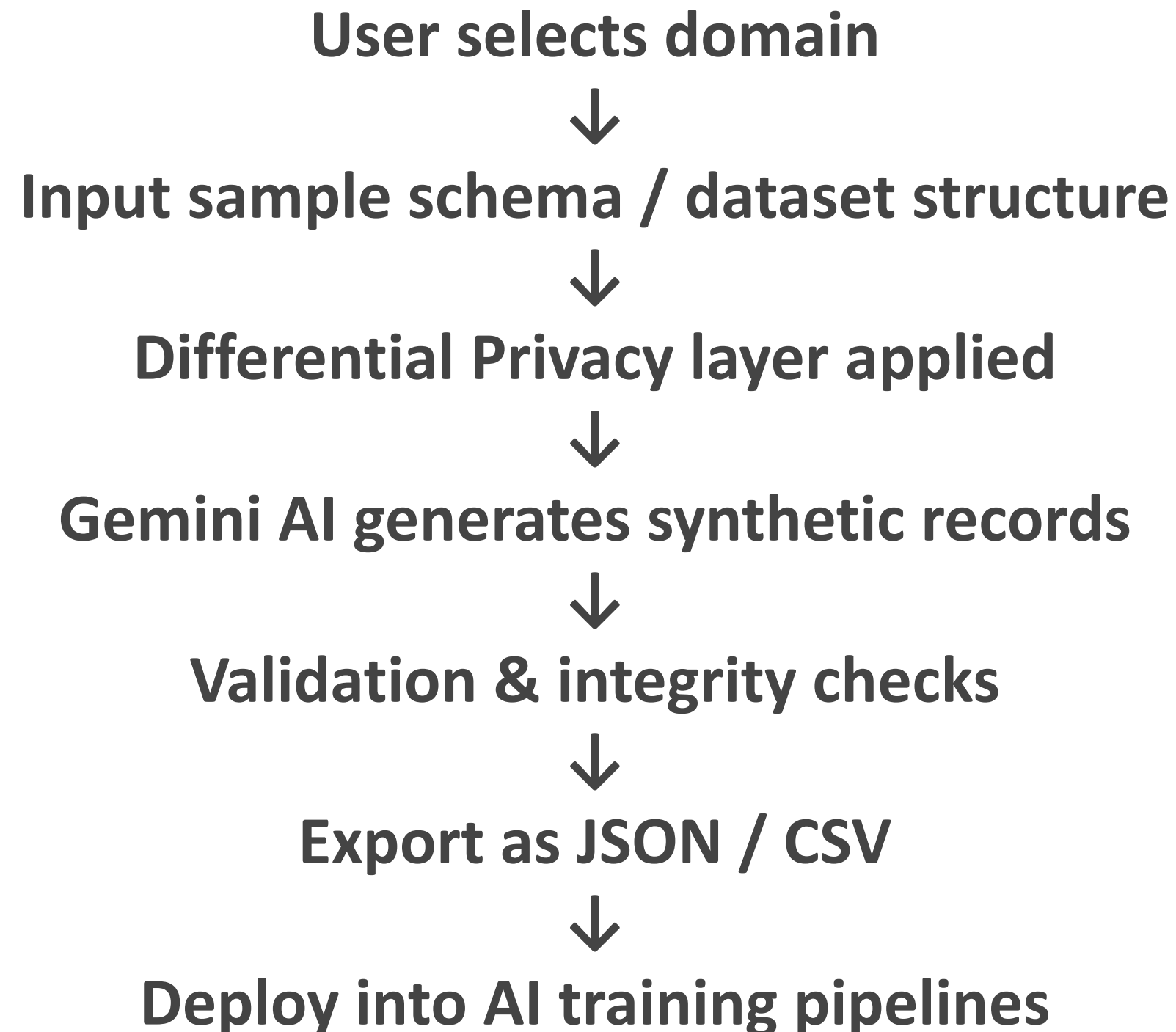
Layer 02 — Persona Synthesis: AI creates vivid fake users (a student with midnight scrolling habits, or an exec with back-to-back calendar blocks).

Layer 03 — Data Tank: populates a virtual device with realistic digital artifacts that match the persona's

Key features

DP-guaranteed privacy · Realistic edge cases · Plugs into Galaxy AI / Knox · Scalable to millions of personas

Demo / Product Walkthrough



- Multi-domain synthetic dataset generation
- Real-time AI-powered record synthesis
- Differential Privacy configuration
- JSON and CSV export support
- AI training readiness scoring
- Synthetic data preview engine

Tech & Architecture

Tech stack

React + Vite + TailwindCSS , Gemini API , Differential Privacy JSON / CSV generation , Schema-aware synthetic synthesis , Vercel-ready architecture

AI usage — model, APIs, framework

Model: claude-sonnet-4 · API: Anthropic /v1/messages · Called twice per run with structured JSON prompts. No fine-tuning. No training data needed.

System flow (simple)

User sets persona + ϵ → DP analysis (Layer 1) → Claude API → Persona JSON (Layer 2) → Claude API → Artifacts JSON (Layer 3) → Data Tank renders: emails · calendar · app logs

Security boundary

Fully compliant with Samsung Knox and Zero Trust. Zero real user data ever enters the pipeline.

Impact & Use Cases

Samsung QA

Galaxy AI's 'Suggested Reply' needs 1,000 diverse inboxes. SynthShield generates them in minutes — zero data exposure.

Healthcare AI

A med-app needs to test AI scheduling. Patient calendars are HIPAA-protected. SynthShield generates synthetic patient profiles instead.

Fintech company

Banking AI needs to test fraud detection on spending habits. Real financial data can't leave the vault. SynthShield creates behaviourally accurate synthetic accounts.

Impact & Use Cases

University researcher

HCI study needs 500 diverse phone-usage profiles. SynthShield generates a research-grade dataset overnight — no ethics board issues.

Scale potential

Any org in healthcare, finance, smart devices, or academia that needs AI testing without privacy risk.

Key Features :

- AI-ready synthetic dataset generation
- Multi-domain schema support
- Differential Privacy guarantees
- JSON & CSV export pipeline
- Enterprise-scale simulation support
- Real-time dataset preview
- Synthetic integrity scoring
- Zero real-data exposure

Use Cases:

HEALTHCARE AI :

Generate synthetic patient datasets for diagnostic model training without violating HIPAA regulations.

FINANCE :

Create transaction datasets for fraud detection and financial AI testing safely.

IoT :

Simulate smart-device telemetry and infrastructure analytics.

EDUCATION :

Generate academic datasets for student analytics and institutional AI systems.

Differentiation

Why our solution stands out

1. Behaviourally realistic synthetic generation
2. Differential Privacy guarantees
3. AI-training-ready export pipeline
4. Multi-domain adaptability
5. Real-time dataset synthesis
6. Enterprise deployment readiness

MOAT 1 — Mathematical privacy (ϵ -DP)

Differential Privacy is hard to implement correctly. Competitors would need months to replicate the guarantee — not just the look.

MOAT 2 — Persona depth & emergent realism

Emails, calendar, and app logs for each persona tell the same coherent human story. That coherence is what makes AI testing meaningful.

Idea

Prototype :

Dataset Generation Configuration

Select Domain

 **Student Data**
Academic records, attendance, grades

 **Medical Data**
Patient records, diagnoses, treatments

 **Agriculture Data**
Crop yields, soil conditions, weather

 **Finance Data**
Transactions, balances, fraud detection

 **Retail Data**
Purchases, customer behavior, inventory

 **IoT / Smart Device Data**
Sensor readings, device telemetry

 **Custom Domain**
Define your own data structure

The types of data that can be generated

Number of Records: 1,000

10 records 10,000 records

Differential Privacy ϵ : 1.0 (Balanced)

Stronger Privacy Higher Utility

 **Generate Synthetic Dataset**

The number and privacy of the data sets

Generated Synthetic Dataset

[JSON](#) [CSV](#) [Copy](#)

 **1,000**
Total Records

 **0.98**
Privacy Score

 **0.95**
Data Integrity

 **Ready**
AI Training Status

Data Preview (First 10 Records)

name	attendance	cgpa	department	semester
Zara Ahmed	78.64	7.85	Computer Science	1
Isha Kapoor	80.98	8.25	Chemical	4
Ananya Reddy	77.31	7.21	Electrical	7
Neha Gupta	93.17	8.52	Civil	2
Rohan Mishra	91.17	8.44	Chemical	5
Rohan Mishra	87.79	7.94	Computer Science	8
Sneha Desai	77.89	7.36	Civil	2
Maya Joshi	88.91	8.92	Computer Science	3
Priya Sharma	82.69	7.38	Biotech	4
Dev Chopra	79.44	8.72	Chemical	1

The data can be downloaded as a Jason file ,
Csv file or the text can be copied as well

Idea

Demo Link

GitHub Repo

github.com/synthshield/artificial-user

Demo Link

<https://musical-phoenix-5c3a6a.netlify.app/>

Document submitted

SynthShield_RV_College_Of_Engineering.pdf

Brownie Slide

SynthShield does not simply randomize records.

It generates statistically coherent synthetic intelligence suitable for real AI workflows.

Today we test Samsung Galaxy AI. Tomorrow: hospital AI, autonomous vehicles, financial advisors — anywhere AI needs realistic human behaviour without privacy risk.

The same 3-layer framework works for any domain.

The personas change. The privacy guarantee doesn't.

Privacy should not slow down AI innovation.

SynthShield enables organizations to safely train, test, and validate AI systems using realistic synthetic intelligence — without exposing sensitive real-world data.